

- 4 A continuous random variable X has cumulative distribution function F given by

$$F(x) = \begin{cases} 0 & x < 1, \\ \frac{1}{5}x + a & 1 \leq x < 4, \\ \frac{1}{50}x^2 + b & 4 \leq x \leq 6, \\ 1 & x > 6, \end{cases}$$

where a and b are constants.

- (a) Find the value of a and the value of b .

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- (b) Find the probability density function of X .

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(c) Given that $E(X) = \frac{529}{150}$, find $\text{Var}(X)$.

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(d) Find the 10th and 90th percentiles of X .

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